

Chapter 1.6 – Ethical, Legal, Cultural and Environmental Impacts of Digital Technology

PART 1: TOPIC NOTES

The Data Protection Act (2018)

Controls stored data about individuals and gives them the right to access, check and correct it. The data controller in an organisation is responsible for accuracy and security of data.

The 8 provisions: data must be processed fairly and lawfully, only used for the specified purpose, relevant and not excessive, accurate and up to date, only kept as long as necessary, individuals can access and correct their data, security must prevent unauthorised access, and data cannot be transferred outside the EU unless that country has adequate protection.

Exemptions: national security, crime and taxation, and domestic/personal use.

The Computer Misuse Act (1990)

It is a criminal offence to make any unauthorised access to computer material with intent to commit further offences (e.g. hacking) or to modify computer material (e.g. distributing malware).

Protection methods include passwords and user IDs, firewalls, anti-malware software, SSL encryption, and digital signatures.

The Copyright Designs and Patents Act (1988)

Protects the intellectual property of individuals or organisations. It is illegal to copy, modify or distribute software or other intellectual property without permission. This includes video and audio content. Most software is issued with a licence explaining what can and cannot be done with it.

The Freedom of Information Act (2000)

Gives the public the right to access information held by public authorities such as government departments, local authorities, schools and the police. Journalists often use this act to scrutinise public authorities.

Software Licences

Open source — source code is freely available, can be modified and redistributed, free or very low cost, developed and tested by a community, but may lack support and full testing.

Proprietary — source code is hidden, cannot be modified or distributed, usually costs money, fully tested and supported by the organisation, copyright protected.

Ethical Issues

Ethics is about what society considers right or wrong. Key ethical concerns in computing include privacy, surveillance, AI decision-making, and the impact of technology on employment and society.

CCTV and facial recognition can improve security and solve crimes but raise concerns about constant surveillance and “big brother” monitoring. Mobile tracking can help police and locate people but records location data and exposes images to hackers. Workplace monitoring helps employers track productivity but intrudes on employees’ privacy. Social media enables communication but enables trolling, cyberbullying, and exposure of private information to wide audiences.

Cultural Issues

Stakeholders are those with an interest in a situation — what is positive for one stakeholder may be negative for another.

Internet censorship suppresses unacceptable or dangerous material through blacklists, whitelists, and dynamic filtering. The debate is between protecting the public and protecting free speech.

Computers in the workforce have changed the skills required — for example robots replacing welders, automated warehouses replacing shop assistants, and online banking reducing the need for high street branches. This creates job losses but also new technical roles.

Environmental Issues

Data centres account for around 2% of all global energy use. A lot of energy and non-renewable resources are used in manufacturing computers. Computers contain toxic materials including mercury, cadmium and chromium. Old equipment is often shipped to countries with lower environmental standards for disposal, sometimes putting children at risk. E-waste leads to toxic substances being released into land, water and air.

PART 2: MARKING SCHEME QUESTIONS AND ANSWERS

Q: Identify which act is broken for each of the following scenarios.

A company transmits personal data to another company without the individual's permission — The Data Protection Act (2018).

A school publishes their student's addresses on the school website — The Data Protection Act (2018).

The interface for a piece of software is replicated by a rival company — Copyright Designs and Patents Act (1988).

A user leaves a computer logged on and another person leaves them a message on their desktop — Computer Misuse Act (1990).

A student guesses their teacher's password and accesses their computer account — Computer Misuse Act (1990).

Using a picture for the law company's new logo without the original creator's permission — Copyright Designs and Patents Act (1988).

A secretary accessing a lawyer's personal email account without permission — Computer Misuse Act (1990).

Making a copy of the latest Hollywood blockbuster movie and sharing it with a client — Copyright Designs and Patents Act (1988).

Storing customer data insecurely — The Data Protection Act (2018).

A lawyer installing a key logger on the secretary's computer — Computer Misuse Act (1990).

Selling client's personal data to a marketing company without their permission — The Data Protection Act (2018).

Q: Describe two benefits to the artist and programmer of releasing the software as proprietary.

Authors can earn money by selling it for a fee / using licences to stop unauthorised use. No-one can see the code. Users cannot edit/change the software, for example by inserting malware. It cannot be copied/resold/shared. More control over intellectual property, for example restricting users and restricting what can be done with the software without permission.

Q: A computer programmer has developed a computer game they want to release for users to download over the internet. Discuss the features, benefits and drawbacks of each type of licence and make a recommendation.

Licence features — Open source is usually free, can access/change source code, redistribute. Proprietary requires purchase at a cost and users cannot access/change code.

Legal and ethical issues — Both provide copyright. Open source allows more people to take code and possibly change to resell or adapt in their own programs to resell or claim as their own (reverse for proprietary). Open source allows more people access to the game because there is likely no cost (reverse for proprietary).

Benefits and drawbacks — Open source gives a wider customer base, more exposure, and users can alter to make it better. Proprietary allows the programmer to earn money and gives more control over what happens with the program; it is usually well tested. However, proprietary is more restrictive for copyright, cannot be adapted to meet user needs, may have bugs, limited documentation, and little financial gain for open source.

Decision — Either would be appropriate. Justification needs to be clearly related to the scenario.

Q: Fiona releases her game under a proprietary licence. Explain why a proprietary licence is a more appropriate choice than open source.

She can sell it for a fee. It protects/copyrights her source code. So it can't be copied/modified/redistributed.

Q: Social networking websites use AI to monitor posts from users. Discuss the positive and negative uses of AI by social networking websites including legal issues, ethical issues and privacy issues.

Legal issues — Copyright Designs and Patents Act can check for plagiarism automatically and highlight posts e.g. videos or images. Data Protection Act needs to make sure rules are followed so that the AI algorithm does not breach e.g. security. Check that materials are all legal. User has agreed to the terms when signing up so should expect it.

Ethical issues — AI can be used to protect users from harmful content and flag inappropriate posts. However, AI may make incorrect decisions and wrongly remove content. There are concerns about who controls the AI and what bias it may have.

Privacy issues — Users may be unaware they are being recorded and not be informed. Users do not know where videos/data about them and their movements is stored/how it is used — DPA evidence. Users have not given consent. Users may feel unsafe because they are being watched. The AI stores and processes personal data which must comply with the Data Protection Act.

Q: A shopping centre has a security system with CCTV cameras and is upgrading to include facial recognition to track and record individuals. Discuss the positive and negative impacts including ethical, privacy and legal issues.

Legal issues — DPA needs to be followed or the company could be fined. Customers must be informed the system is used. Data must be held for specified time/reasons. Data must be kept secure. The centre is private property so customers can choose not to enter. Customers informed the system is used.

Ethical issues — Users feel safer because they know any actions are being monitored and help/action will be taken if needed. If users have not done anything then there is no reason to be tracked/recorded so should not impact them. Users feel unsafe because they are being watched all the time. Users may be unaware they are being recorded and not

be informed. Users do not know where videos/data about them and their movements is stored/how it is used. Users have not given consent.

Privacy issues — Users are in a public place and can be legally recorded by anyone. Users may feel like they are being watched all the time. Users have not given consent. Who is responsible if there is an error.

Q: Daniel is a medical researcher trying to find a cure for a disease. He decides to increase his use of AI instead of hundreds of researchers. Discuss the issues surrounding this decision including ethical, legal and cultural issues.

Ethical issues — Replacing people with machines. Loss of jobs. Community will suffer. Work will be completed faster. May find a cure faster. More reliable calculations. Save more lives.

Legal issues — More secure than people seeing personal data. May be at risk if not backed up. May be at risk of threats e.g. hackers. Who is responsible if there is an error.

Cultural issues — Removal of people from workforce. Change in demand for skills. Need people to manage the hardware/software instead of medical expertise. Skills may be lost.

Q: People often want to buy the most up-to-date smartphones even though their own still works. Discuss the impact of people wanting to upgrade to the latest smartphone.

Smartphone users — Can adversely affect people in this country and abroad including health issues, financially, socially, and culturally.

Cultural issues — Desire/need to own newest device. To fit in with peers. May have new features that users require for work/leisure.

Environmental issues — The type of devices that are disposed of. Modern phones poorly designed for durability. Phones' hardware not upgradeable/replaceable. Reference to e-waste — people dispose of their devices in landfill even if they are in good working order. Some equipment is also sent abroad to be disposed of. Leads to excessive landfill in this country and/or abroad e.g. Africa and Asia. Toxic waste released into land, ground water, air in this country and/or abroad e.g. Africa and Asia. Waste of resources. Precious metals in phones.

Ethical issues — Contributes to ill health. Contributes to the digital divide. Contributes to social divide. Problem of confidential data stored on the devices. Puts social pressure on parents to pay for their children to upgrade. Puts social pressure on the public to upgrade. Can lead to bullying of those who cannot afford the latest technology. Phone manufacturers intentionally designing fragile phones so they need to be replaced more often. High cost of new devices.

Q: Fiona is selling her game online only and not making it available in shops. Describe the environmental impact of Fiona's decision.

Benefits of not providing physical copies — Less/no plastic/paper/raw materials

used in manufacture / no need for packaging / less waste. Less electrical power needed to manufacture. No petrol used to distribute/collect. Smaller carbon footprint. Fewer disks need to be manufactured. Fewer factory emissions / less pollution. Old versions will be thrown away.

Drawbacks of still creating a physical copy sold online — Plastic/paper are used in manufacture. Increase in waste. Old versions will be thrown away. Uses petrol / creates emissions to distribute.